

Policy: Unique Identification of Medical Records

Policy statement and conformance indicator specification

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POLICY OWNER	Director, Health Information Management (HIM)
TECHNICAL OWNER	Data Governance Lead, Enterprise Master Patient Index (EMPI)
REVIEW CYCLE	Annual, or upon any change to registration, EMPI, or source-system architecture
PRIMARY INDICATOR	DQI-001 — Unique Identifier Integrity Index (UIII)

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Note on use. This document is written as an operational policy with a measurable data-quality indicator. Organisation-specific identifier formats, thresholds, and escalation paths are marked as configurable and must be confirmed by the policy owner. Regulatory citations are informative; legal and compliance review is required before adoption.

1. Purpose

Every medical record must be attributable to exactly one patient, and every patient must be reachable through exactly one primary record identifier. When that correspondence breaks, clinicians treat patients from incomplete or wrong charts, billing is misdirected, and disclosure controls fail.

This policy establishes:

1. The rule that every medical record carries a unique, permanent, machine-verifiable identifier.
2. A quantitative **indicator** that measures conformance to that rule.
3. The thresholds, measurement cadence, ownership, and remediation path that make the indicator actionable rather than decorative.

2. Scope

In scope. All systems that create, store, or exchange patient-level records: the electronic health record (EHR), the EMPI, registration and scheduling, laboratory, radiology/PACS, pharmacy, revenue cycle, the clinical data warehouse, research extracts, and any interface engine that routes patient records between them.

In scope regardless of medium. Electronic records, scanned documents, and legacy paper records indexed into the EMPI.

Out of scope. De-identified datasets that have had identifiers removed under an approved de-identification procedure, and fully synthetic data. These remain subject to their own governance and to Section 3.7 below.

Definition of "the Organization". The covered entity and any affiliated site operating under the shared EMPI.

3. Policy statements

P-1 — Mandatory identifier. Every medical record shall be assigned a Medical Record Number (MRN) at the moment of creation. No record may be saved, transmitted, or billed without one. Null, blank, placeholder ("UNKNOWN", "TEMP", "0000000"), and system-default identifiers are non-conformant except under the controlled unidentified- patient procedure in P-6.

P-2 — Uniqueness. An MRN shall identify exactly one patient within its assigning authority. The database shall enforce this with a `UNIQUE NOT NULL` constraint; the constraint is the control, and the indicator in Section 4 is the verification that the control held everywhere, including in systems the constraint does not reach.

P-3 — Singularity. A patient shall hold exactly one active MRN per assigning authority. Additional identifiers discovered for the same person shall be merged or linked through the EMPI, never left as parallel active records.

P-4 — Permanence and non-reuse. An MRN is permanent for the life of the record and beyond. It shall never be reused for a different patient, including after death, record purge, or account closure. Retired identifiers shall be retained in a tombstone table so that reuse is detectable.

P-5 — Format conformance. MRNs shall conform to the registered format for their assigning authority (*configurable; default: a fixed-length numeric string with a check digit*). Identifiers shall be opaque: they shall not encode name, date of birth, national

identification number, or any other clinical or demographic attribute.

P-6 — Unidentified patients. Where care must begin before identity is established (trauma, unresponsive patient, mass-casualty intake), a temporary identifier shall be issued from a reserved, clearly flagged range. Every temporary identifier shall be reconciled to a permanent MRN within **24 hours** of identification. Open temporary identifiers older than 24 hours are counted as defects by the indicator.

P-7 — Cross-system integrity. Records exchanged between systems shall carry both the local identifier and its assigning authority, so that the pair (`assigning_authority` , `identifier`) is globally unique. Interfaces shall reject inbound records lacking this pair. This aligns with the HL7 FHIR `Patient.identifier` structure of `system` + `value` .

P-8 — Merge and unmerge discipline. Only trained HIM staff may merge or unmerge records. Every merge shall record the surviving MRN, the retired MRN(s), the operator, the timestamp, and the evidence used. Every merge shall be reversible.

P-9 — Measurement. Conformance to P-1 through P-8 shall be measured by indicator DQI-001 on the cadence in Section 5. An unmeasured control is treated as a failed control.

4. Indicator specification — DQI-001

4.1 Header

Attribute	Specification
Indicator ID	DQI-001
Indicator name	Unique Identifier Integrity Index (UIII)
Data quality dimension	Uniqueness / identity integrity (DAMA-DMBOK dimension set)
Type	Composite percentage, derived from three component indicators
Unit	Percent (0.00–100.00), reported to two decimal places
Direction	Higher is better
Population	All active medical records in scope at the measurement instant
Exclusions	Records flagged as test/training data; records in the reserved temporary range that are less than 24 hours old

4.2 Terminology

The three failure modes are distinct and are measured separately, because they carry different clinical risk and different remediation:

Term	Meaning	Risk
Missing	A record with no valid identifier	Record is unfindable; care delivered blind
Overlay	One MRN pointing at more than one patient	Highest — one patient's chart is read as another's
Duplicate	One patient holding more than one active MRN	Fragmented history; allergies and results are split across charts
Overlap	The same patient held under different MRNs in different assigning authorities	Managed by linkage, not merge; tracked separately

4.3 Component indicators

Let the measurement population be **R** = all in-scope active records, and **P** = all distinct patients represented in **R**.

I-1 — Identifier Assignment Rate

$$I-1 = (| \text{ records in R with a non-null, format-conformant MRN } | / |R|) \times 100$$

Measures P-1 and P-5. A record fails I-1 if the MRN is null, blank, a placeholder value, fails the registered format or check digit, or is an unreconciled temporary identifier older than 24 hours.

I-2 — Overlay-Free Rate

$$I-2 = (1 - (| \text{ records whose MRN is shared by more than one distinct patient } | / |R_{id}|)) \times 100$$

where **R_id** is the subset of **R** holding an MRN. Measures P-2. Note that both sides of an overlay count as defects: if one MRN maps to two patients, two records are defective.

I-3 — Duplicate-Free Rate

$$I-3 = (| \text{ patients in P holding exactly one active MRN } | / |P|) \times 100$$

Measures P-3. Patient identity for this calculation is resolved by the EMPI matching algorithm at its configured auto-link confidence threshold; records in the manual review queue are counted as duplicates until cleared, so that an unworked queue degrades the score rather than hiding in it.

4.4 Composite

$$UIII = (0.25 \times I-1) + (0.45 \times I-2) + (0.30 \times I-3)$$

Overlay carries the heaviest weight because it is the only failure mode that can place one patient's clinical data in front of a clinician treating another. Weights are configurable by the policy owner but shall keep **w(I-2)** as the largest term.

4.5 Worked example

A daily run over 1,000,000 active records covering 940,000 distinct patients finds:

- 1,200 records with no usable MRN (missing or malformed, so excluded from **R_id**) → $I-1 = (998,800 / 1,000,000) \times 100 = \mathbf{99.88}$
- 6 MRNs each shared by 2 patients → 12 defective records out of 998,800 → $I-2 = (1 - 12/998,800) \times 100 = \mathbf{99.9988}$
- 3,100 patients holding more than one active MRN → $I-3 = (936,900 / 940,000) \times 100 = \mathbf{99.67}$

$$\begin{aligned} UIII &= (0.25 \times 99.88) + (0.45 \times 99.9988) + (0.30 \times 99.67) \\ &= 24.9700 + 44.9995 + 29.9010 \\ &= 99.87 \end{aligned}$$

UIII of 99.87 rates **Green** on the scale below — but the 6 overlays independently trigger a Severity-1 response under Section 4.6. The composite never suppresses an overlay finding.

4.6 Thresholds and rating

Rating	UIII	Response
Green	≥ 99.50	Routine monitoring
Amber	98.00 – 99.49	Root-cause review within 5 business days; corrective plan to the Data Governance Committee
Red	< 98.00	Escalation to the Chief Medical Information Officer within 24 hours; registration pathway review

Overriding rule — overlays. The target for overlays is **zero**. Any confirmed overlay is a Severity-1 patient-safety event regardless of the composite score: the affected records are quarantined from clinical view within 1 hour of confirmation, the attending clinicians and the Patient Safety Officer are notified, and a root-cause analysis is filed within 5 business days.

Secondary targets. Duplicate rate $(100 - I-3)$ shall not exceed **2.0%**, consistent with widely used industry benchmarks for EMPI duplicate rates; the policy owner may tighten this. Temporary identifiers unreconciled beyond 24 hours shall not exceed 0.1% of records created in the period.

5. Measurement procedure

5.1 Cadence

Layer	Control	Frequency
Preventive	UNIQUE NOT NULL constraint; check-digit validation at entry; duplicate-search prompt before new-record creation	Real time, at registration
Detective	Automated DQI-001 calculation over the clinical data warehouse	Daily, 02:00 local
Detective	Full cross-authority overlap scan	Weekly
Reporting	Indicator pack to the Data Governance Committee	Monthly
Assurance	Independent audit of the calculation logic and of a sample of merges	Annual

5.2 Reference queries

The measurement job is the authoritative implementation; the queries below define its intended semantics and are written in ANSI SQL against a conformed view `medical_record` (`record_id`, `mrn`, `assigning_authority`, `patient_key`, `status`, `created_at`, `is_test`).

Missing or malformed identifiers (I-1 defects):

```
SELECT COUNT(*) AS missing_or_malformed
FROM   medical_record
WHERE  status = 'ACTIVE'
      AND is_test = FALSE
      AND (mrn IS NULL
          OR TRIM(mrn) = ''
          OR mrn IN ('UNKNOWN', 'TEMP', '0000000')
          OR mrn NOT SIMILAR TO '[0-9]{8}'); -- registered format; configurable
```

Overlays — one identifier, several patients (I-2 defects):

```

SELECT    assigning_authority,
          mrn,
          COUNT(DISTINCT patient_key) AS patients_sharing_mrn,
          COUNT(*)                    AS defective_records
FROM      medical_record
WHERE     status = 'ACTIVE' AND is_test = FALSE AND mrn IS NOT NULL
GROUP BY  assigning_authority, mrn
HAVING    COUNT(DISTINCT patient_key) > 1
ORDER BY  patients_sharing_mrn DESC;

```

Duplicates — one patient, several identifiers (I-3 defects):

```

SELECT    patient_key,
          COUNT(DISTINCT mrn) AS active_mrn
FROM      medical_record
WHERE     status = 'ACTIVE' AND is_test = FALSE AND mrn IS NOT NULL
GROUP BY  patient_key
HAVING    COUNT(DISTINCT mrn) > 1;

```

Reuse of a retired identifier (P-4 breach; always Severity-1):

```

SELECT r.mrn, r.retired_at, r.retired_patient_key, m.patient_key AS reissued_to
FROM    retired_mrn r
JOIN    medical_record m
      ON m.mrn = r.mrn
      AND m.assigning_authority = r.assigning_authority
WHERE   m.patient_key <> r.retired_patient_key;

```

5.3 Evidence retention

Each run shall persist the component values, the composite, the population size, the defect record keys, and the code version that produced them. Results are retained for **7 years** (*configurable to the Organization's record-retention schedule*) so that the indicator series is itself auditable.

6. Reporting

The monthly pack shall present:

- UIII as a **run chart** over at least 13 periods, with the Green/Amber threshold bands drawn on the chart. A single-period number invites over-reaction to noise; the series shows whether the process is stable.
- The three components as separate series — a flat composite can hide I-2 degrading while I-1 improves.
- Overlay count as an absolute count, never a percentage. Percentages of very small numbers read as zero and are how overlays get ignored.
- Defect volume by originating system and by registration point, to direct remediation at the source rather than at the symptom.

7. Roles and responsibilities

Role	Responsibility
Data Governance Committee	Approves this policy, the thresholds, and the weights; reviews the monthly pack; adjudicates exceptions
Director, HIM (policy owner)	Owns the policy and the indicator definition; owns merge/unmerge procedure and staff training
Data Governance Lead / EMPI (technical owner)	Owns the measurement job, its correctness, and its scheduled execution; publishes the indicator
Registration and Patient Access	First line of prevention: duplicate search before record creation; correct capture of demographics
Application owners	Enforce the database constraint and interface validation in their system; remediate defects attributed to it
Patient Safety Officer	Co-leads the response to any overlay
Internal Audit	Annual independent verification of calculation logic and merge sampling

8. Remediation

1. **Detect.** The daily job writes defects to the data-quality work queue, classified as missing, overlay, duplicate, overlap, or reuse.
2. **Triage.** Overlays and reuse are Severity-1 and jump the queue. Missing identifiers on records with clinical activity in the last 30 days are Severity-2. Everything else is Severity-3.
3. **Contain.** For an overlay, quarantine the affected records from clinical view within 1 hour of confirmation and notify the treating clinicians.
4. **Correct.** HIM performs the merge, unmerge, or identifier assignment under P-8, with full audit trail.
5. **Verify.** The next scheduled run must show the defect cleared. Defects that reappear are escalated as a control failure, not re-worked as a new defect.
6. **Prevent.** Any Severity-1, and any Amber or Red period, requires a root-cause analysis naming the system or process that admitted the defect, and a dated corrective action.

9. Exceptions

Exceptions shall be requested in writing to the Data Governance Committee, shall state a business justification, a compensating control, and an expiry date, and shall not exceed 12 months. No exception may be granted to P-2 (uniqueness) or P-4 (non-reuse). Approved exceptions are recorded in the exception register and their affected records are still counted as defects by DQI-001 — an exception permits the condition, it does not make the measurement lie.

10. Non-compliance

Systems that cannot demonstrate conformance to P-1, P-2, and P-7 shall not be granted or shall lose approval to exchange patient records across the interface engine. Repeated individual non-compliance with the merge procedure in P-8 is handled under the Organization's existing disciplinary and retraining process.

11. Definitions

Term	Definition
Assigning authority	The organisation or system namespace responsible for issuing a given identifier; makes an identifier globally unique when paired with the identifier value
Check digit	A digit computed from the other digits of an identifier, used to detect transcription error at entry
EMPI	Enterprise Master Patient Index — the service that resolves records across systems to a single patient identity
Medical record	Any record holding patient-level clinical or administrative information about an identifiable individual
MRN	Medical Record Number — the primary identifier issued by an assigning authority
Tombstone record	A retained stub for a retired identifier, kept so that reuse is detectable
UIII	Unique Identifier Integrity Index — the composite indicator DQI-001

12. Informative references

These informed the policy; they are listed for orientation and are not a compliance determination. Confirm current applicability with the compliance function.

- HIPAA Security Rule, integrity standard — 45 CFR § 164.312(c)(1)
- HIPAA Privacy Rule, amendment of protected health information — 45 CFR § 164.526
- The Joint Commission, National Patient Safety Goal on patient identification (NPSG.01.01.01)
- DAMA-DMBOK, data quality dimensions (uniqueness)
- ISO/IEC 25012, data quality model
- HL7 FHIR, `Patient.identifier ( system + value )` and identifier use codes

13. Revision history

Version	Date	Author	Change
1.0	2026-09-22	Data Governance	Initial policy and DQI-001 indicator specification